Impact of Social Media on Student Life

1. Project Overview and Objective

This Project involves cleaning,transformation,and analyzing raw student Social Media data using Excel and creating an interactive Power BI dashboard to drive meaning insights.

The main objectives is demonstrates data per-processing techniques using Excel and Power BI dashboard visualization to understand how social media affects students' sleep, stress, and GPA, and to make informed decisions for students wellness.

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2. Data Sources:

- **Source Description : Student Social Media Dataset from Kaggle.**
- **Domain : Education / Psychology / Social Media Analytics** etc

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3. Problem Statement :

- To analyse social media usage patterns across Age, Gender, Acadamic Level to identity high-usage groups.
- To study the relationship between Daily Usage Hours and sleep Duration, sleep Quality and Mental Health Index.
- To evaluate the impact of Late-Night Usage and Social Comparison on Preceived Stress Score.
- To understand how Primary Platform (TikTok, Instagram, YouTube, LinkedIn, Snap chat, X(Twitter), Reddit) Influences Acadamic Performance (GPA) and Overall Impact.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
STudent_ID	Text	Unique identifier for each student
Age	Numeric	Age of Student in year
Gender	Categorical	Female / Male / Non-binary / Prefer not to say
Academic_level	Categorical	High School / Undergraduate / Postgraduate
Primary_Platform	Categorical	TikTok / Instagra/ YouTube / LinkedI / Snap chat / X(Twitter) /Reddit

Daily_Usage_Hours	Numeric	Average daily social media usage
Weekend_Extra_Hours	Numeric	Average Extra hours usage in Weekend
Device Type	Text	Smart Phone / Laptop / PC / Tablet
Sleep_Duration_Hours	Numeric	Average sleep hours
Sleep_Quality_Score	Numeric	Average sleep quality score
Late_Night_Usage	Categorical	Ture / False
Preceived _Stress_Score	Numeric	Average stress score
Mental_Health_Stress	Numeric	Mental Health score(1-10)
Academic_Performance _GPA	Numeric	GPA Score
Overall_Impact	Categorical	Positive / Negative

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query) (Sample)

Tasks Performed:

- **Data Cleaning & Transformation:**
- Removed duplicates
- Handled missing value:

Using IF(OR(ISBLANK(M2),M2=0),AVERAGE(M2:M4501),M2) formula to fill the blank value of Preceived_stress_score

Using IF(OR(ISBLANK(O2),O2=0),AVERAGE(O2:O4501),O2)f formula to fill the blank value of Academic_performance_GPA

- **standardized formats, and created calculated fields.**

1. =IF(M3>20,"HIGHSTRESS",IF(M3>10,"MEDIUMSTRESS","LOW STRESS"))

2. =IF(O2<2,"LOW GPA",IF(O2<3,"MEDIUM GPA ","HIGH GPA"))

- **Filtering & Sorting: Organized data to focus on relevant records.**

Social_Comparison_Frequency	Perceived_Stress_Score	Mental_Health_Index	Academic_Performance_GPA	Overall_Impact	Stress_Level	GPA_PERFORMANCE
Always	40.00	38.00	1.99	Negative	HIGH STRESS	LOW GPA
Rarely	34.00	36.00	1.97	Negative	HIGH STRESS	LOW GPA
Sometimes	36.00	38.00	1.97	Negative	HIGH STRESS	LOW GPA
Never	35.00	39.00	1.90	Negative	HIGH STRESS	LOW GPA
Sometimes	36.00	43.00	1.95	Negative	HIGH STRESS	LOW GPA
Sometimes	35.00	45.00	1.99	Negative	HIGH STRESS	LOW GPA

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Sometimes	36.00	38.00	1.97	Negative	HIGH STRESS
Never	35.00	39.00	1.90	Negative	HIGH STRESS
Sometimes	36.00	43.00	1.95	Negative	HIGH STRESS
Sometimes	35.00	45.00	1.99	Negative	HIGH STRESS

- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

Gender	Average of Academic_Performance_GPA
Female	3.422
Male	3.435
Non-Binary	3.467
Prefer not to say	3.425
Grand Total	**3.429**

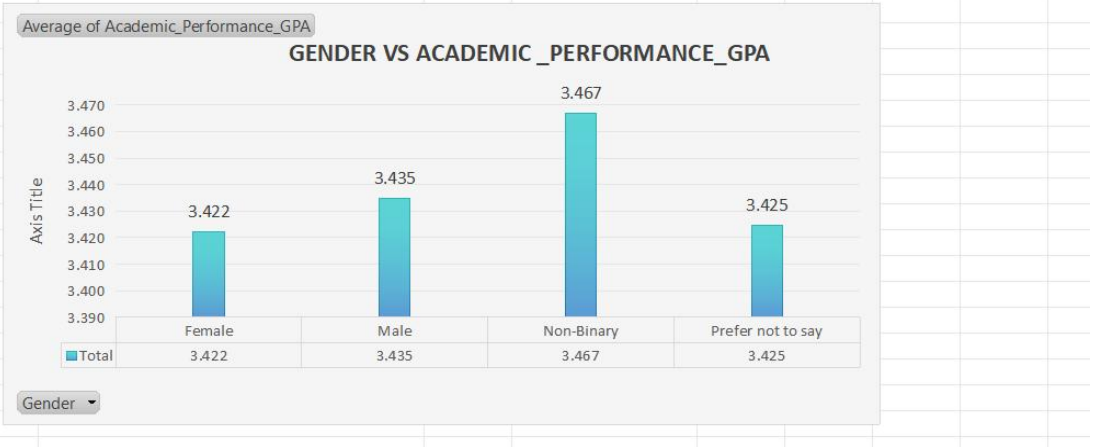
DIAGNOSIS

***The average score is 3.47 Non- Binary has the highest score**

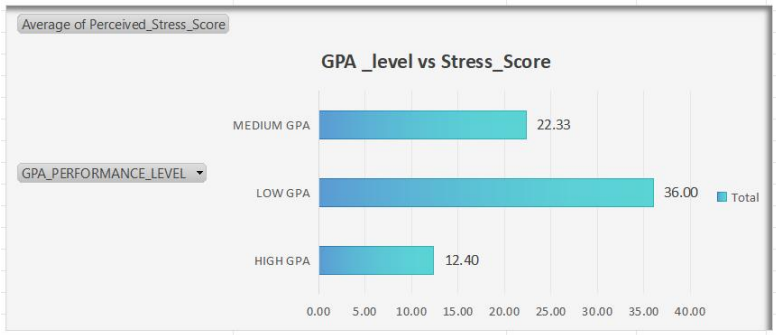
***Female has the lowest score of 3.422**

***The next lowest score of 3.425 prefer not to say**

*** But the range between all the genders are narrow range of variation.**



GPA_PERFORMANCE_LEVEL	Average of Perceived_Stress_Score	DIAGNOSIS
HIGH GPA	12.40	*The bar chart displays the Total for different GPA levels. LOW GPA has the highest Total at 36.
LOW GPA	36.00	*HIGH GPA has the lowest stress level 12.3981
MEDIUM GPA	22.33	*The students who have Low level GPA they have heighest level of stress
Grand Total	**13.78**	



- Convert the data into Fact and Dimension Table **(If Possible)**

=VLOOKUP('Social_media_impact_on_life (1)'!A2,A:D,4,FALSE)

=VLOOKUP('Social_media_impact_on_life (1)'!A2,A:D,2,FALSE)

=VLOOKUP('Social_media_impact_on_life (1)'!A2,A:D,3,FALSE)

Using vlookup to create fact and dimension table.

FactTable: Student_ID

Academic_Performance_GPA

Perceived_Stress_Score

Mental_Health_Index

Sleep_Duration

DIM_GPA:

Academic_Performance_GPA

GPA Level

DIM_Stress_level

Perceived_Stress_Score

Stres level

DIM_Students:

Students_ID

Age

Gender

Academic Level

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING

- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**. (List Out)**

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**
- Make your Report interactive Layout using **Drill-down**, **Filters and slicers**
- **Bookmarks, Clear Titles for each visuals & Labels is must.**
- **Create a consolidated Report /Dashboard.**

SCREENSHOT OF DASHBOARD /REPORT

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9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data
- **Provide the analysis insights:**
  - **Descriptive**
  - **Diagnosis**
  - **Predictive**
  - **Prescriptive**

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10. Conclusions (Sample)

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting.